# Project Timeline & Role-Wise Work Plan

## ML-Based Trust-Aware Validator Selection for Blockchain Consensus

**Start date: Tuesday, September 8, 2026 Total estimated duration: ~18 days → ends Friday, September 25, 2026**

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## Overview Table (Sequence of Work)

| Phase | Role | Task | Duration | Days | Dates |
|---|---|---|---|---|---|
| 1 | Role 4 | Research everything | 3 days | Day 1–3 | Sep 8 (Tue) – Sep 10 (Thu) |
| 2 | Role 1 | Simulate blockchain + collect 1000-transaction behavior data | 4 days | Day 4–7 | Sep 11 (Fri) – Sep 14 (Mon) |
| 3 | Role 2 | Build & train ML model, generate trust scores | 3 days | Day 8–10 | Sep 15 (Tue) – Sep 17 (Thu) |
| 4 | Role 1 | Run both comparison rounds (100 transactions each) | 2 days | Day 11–12 | Sep 18 (Fri) – Sep 19 (Sat) |
| 5 | Role 3 | Compare datasets, write final report/conclusion | 3 days | Day 13–15 | Sep 20 (Sun) – Sep 22 (Tue) |
| 6 | All Team | Integration check, common testing, polish, presentation prep | 3 days | Day 16–18 | Sep 23 (Wed) – Sep 25 (Fri) |

> **Note:** Day 6, 12, and 13 fall on a Saturday/Sunday. If your team doesn't work weekends, add 2 extra days and the project will finish around **Sep 27–28** instead. Adjust as needed for your actual availability.

Work is mostly **sequential** (each phase depends on the previous one's output) — but Roles 3 and 4 have downtime early/late that they should use productively (see notes below).

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## PHASE 1 — Role 4 (Research) | Day 1 to Day 3 (Sep 8 – Sep 10)

**Goal:** Give the whole team a clear foundation before anyone writes code.

### Day 1 (Tue, Sep 8): Blockchain & Consensus Basics

- Study how blockchain works — nodes, blocks, hashing, decentralization
- Study Proof-of-Work vs Proof-of-Stake in detail
- Understand exactly how traditional PoS validator selection works (the stake-based probability formula)
- Study the Ronin Network hack (2022) as the real-world motivating case — note it was Proof-of-Authority (PoA), not pure PoS, and be precise about this in the report

### Day 2 (Wed, Sep 9): Trust/Reputation-Based Systems + Feature Identification

- Research existing trust-based/reputation-based consensus approaches (StakeDag, PoIR, reputation-based DPoS, etc.) — just enough to summarize, not deep technical replication
- **Most important task of this day:** finalize the exact behavioral factors the project will track — response time/uptime, transaction verification correctness, transaction approval/decline outcome, suspicious activity flag
- Document *why* each factor matters (this becomes your feature justification in the report)

### Day 3 (Thu, Sep 10): Compile Findings + Brief the Team

- Write up the research findings into a clean document (introduction, problem statement, literature summary, chosen features list)
- **Share the finalized feature list with Role 1 by end of Day 3** — Role 1 cannot start meaningfully without this
- Note the fintech angle (P2P lending / payment settlement) and how it justifies the project's importance

**Deliverable by end of Day 3:** Research document + finalized list of behavioral features to simulate.

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## PHASE 2 — Role 1 (Blockchain Simulation + Data Collection) | Day 4 to Day 7 (Sep 11 – Sep 14)

**Goal:** Build the simulated blockchain and produce the 1000-transaction training dataset.

### Day 4 (Fri, Sep 11): Set Up Validator Pool

- Create 100 simulated validators, each with a randomly assigned stake value
- Assign each validator a hidden "true nature" — honest or malicious — used only for later evaluation, not visible to the selection logic
- Design the random behavior generator: honest validators get better/more consistent stats, malicious ones get noisier/worse stats (with realistic overlap, not too obviously separable)

### Day 5 (Sat, Sep 12): Build the Transaction Simulation Loop

- Write the core loop that runs transactions one at a time
- For each transaction: pick a validator (for data-generation purposes, this can just be random/stake-based at this stage — the real comparison logic comes later in Phase 4), simulate its behavior for that round (uptime status, correct/incorrect verification, approve/decline outcome)
- Run this loop for **1000 transactions**

### Day 6 (Sun, Sep 13): Data Recording

- Log every transaction's data in a structured format (e.g., CSV): `validator_id, transaction_number, uptime, verification_correct, transaction_outcome, stake, true_label`
- Clean and double-check the dataset — no missing values, realistic value ranges, correct proportion of honest vs malicious behavior

### Day 7 (Mon, Sep 14): Finalize & Handoff

- Review the dataset for quality (e.g., plot a few distributions to sanity-check honest vs malicious separation isn't too extreme or too weak)
- **Hand off the 1000-transaction dataset to Role 2**
- Use any spare time to start drafting the code structure for Phase 4 (the two comparison-run selection logic) so it's ready to plug in once the trust score model exists

**Deliverable by end of Day 7:** Clean 1000-transaction behavior dataset for 100 validators, handed to Role 2.

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## PHASE 3 — Role 2 (ML Model Development) | Day 8 to Day 10 (Sep 15 – Sep 17)

**Goal:** Turn the behavior dataset into a working trust-score model.

### Day 8 (Tue, Sep 15): Data Prep + First Model

- Load Role 1's dataset, do feature engineering if needed (e.g., aggregate stats per validator: average uptime, verification accuracy rate, etc.)
- Handle class imbalance (malicious validators should be a minority — use balanced sampling or class weighting if needed)
- Train a first-pass Random Forest classifier/regressor on this data

### Day 9 (Wed, Sep 16): Tune & Evaluate

- Tune hyperparameters (number of trees, depth, etc.)
- Evaluate using precision, recall, F1-score against the known ground-truth labels
- Extract feature importance (which behavior signals matter most — good for your report's interpretability section)
- Decide the cold-start rule (what score a brand-new validator with no history gets)

### Day 10 (Thu, Sep 17): Package the Model

- Wrap the trained model into a clean, reusable function: `get_trust_score(validator_behavior_data) → score (0 to 1)`
- Test this function on a few sample inputs to confirm it works correctly and predictably
- **Hand off the trained model/function to Role 1**

**Deliverable by end of Day 10:** Trained trust-score model, packaged as a callable function, handed to Role 1.

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# PHASE 4 — Role 1 (Run Comparison Rounds) | Day 11 to Day 12 (Sep 18 – Sep 19)

**Goal:** Generate the two datasets that will actually be compared.

## Day 11 (Fri, Sep 18): Build Both Selection Methods

- Implement traditional selection: `Selection Weight = Stake`
- Implement trust-based selection: `Selection Weight = Stake × Trust Score`, calling Role 2's `get_trust_score()` function
- Make sure both methods run on the **exact same underlying validator behavior sequence** for the 100 test transactions — only the *selection method* should differ between the two runs, so the comparison is fair

## Day 12 (Sat, Sep 19): Run Both Rounds + Update Trust Scores Live

- **Run A:** 100 transactions using traditional stake-only selection — record which validator gets picked each time
- **Run B:** 100 transactions using trust-weighted selection — after each transaction, refresh the validator's recent-behavior stats and re-query `get_trust_score()` so the score reflects up-to-date behavior (the model itself isn't retrained, just its input refreshes)
- Save both datasets (Run A results, Run B results) clearly labeled
- **Hand off both datasets to Role 3**

**Deliverable by end of Day 12:** Two labeled comparison datasets (traditional vs. trust-based, 100 transactions each), handed to Role 3.

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# PHASE 5 — Role 3 (Comparison & Final Report) | Day 13 to Day 15 (Sep 20 – Sep 22)

**Goal:** Analyze the two datasets and produce the project's core conclusion.

## Day 13 (Sun, Sep 20): Analyze Run A (Traditional)

- Cross-reference each selected validator against their true label (honest/malicious)
- Calculate: how many times malicious validators were selected under traditional PoS, and how often

## Day 14 (Mon, Sep 21): Analyze Run B (Trust-Based) + Compare

- Do the same analysis for Run B
- Directly compare the two: e.g., “Validator #47 (malicious) was selected 12 times under traditional PoS but only 2 times under trust-based selection”
- Create simple charts/graphs (bar chart of malicious selections per method, trust score trend over rounds, etc.)

### Day 15 (Tue, Sep 22): Write the Final Report

- Summarize the comparison numerically and visually
- Write the conclusion: does the trust-based method meaningfully reduce malicious validator selection? Any trade-offs observed?
- Compile this into the final report section, ready to merge with Role 4’s research section

**Deliverable by end of Day 15:** Final comparison analysis + written conclusion.

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## PHASE 6 — Whole Team | Day 16 to Day 18 (Sep 23 – Sep 25)

Since you mentioned testing will be done together as a team:

- **Day 16 (Wed, Sep 23):** Everyone runs the full pipeline together end-to-end, checking that all pieces (data → model → selection → comparison) connect correctly. Fix any integration bugs as a team.
- **Day 17 (Thu, Sep 24):** Combine everyone’s individual write-ups (Role 4’s research, Role 1’s methodology, Role 2’s model details, Role 3’s results) into one unified project report/document.
- **Day 18 (Fri, Sep 25):** Prepare the presentation/slides and rehearse explaining the project — make sure everyone can explain the full flow, not just their own role, in case of viva questions.

**Project completion target: Friday, September 25, 2026**

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## Key Handoff Points (Don’t Skip These)

1. **By Sep 10 (Thu):** Role 4 → Role 1 (finalized feature list)
2. **By Sep 14 (Mon):** Role 1 → Role 2 (1000-transaction dataset)
3. **By Sep 17 (Thu):** Role 2 → Role 1 (trained trust-score model/function)
4. **By Sep 19 (Sat):** Role 1 → Role 3 (both comparison datasets)
5. **By Sep 22 (Tue):** Role 3 → Whole team (final results, ready for report merge)

If any handoff is late, every phase after it shifts — so keep communication tight around these dates specifically.